

1 EX-003 — heater/thermocouple layout

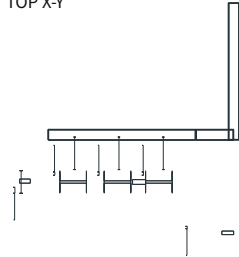
물리 검증·안전 인증·통전 승인이 아니다. E-stop, lid/service interlock, branch fuse, 독립 thermal fuse를 정상 firmware와 독립 구현하고 exact donor 정격·배선·보호소자를 실측 확인하기 전 통전하지 않는다.

Revision: final-design-fabrication-closure-v0.8 · 상태: DIGITAL_DOCUMENT / PHYSICAL_NOT_RUN / USER_APPROVAL_REQUIRED

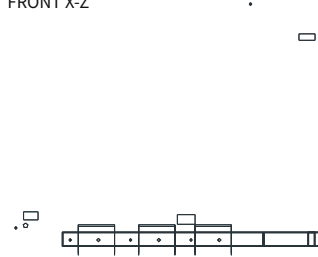
EX-003 heater thermocouple

v0.8 · mm · 제3각법 · source: FreeCAD Python · bodies 18

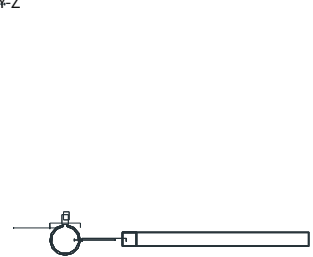
TOP X-Y



FRONT X-Z



SIDE Y-Z



Overall X 382.00 · Y 427.00 · Z 313.50 mm · 치수/공차는 개별 제작도면과 interface catalog 우선
Revision final-design-fabrication-closure-v0.8 · physical validation NOT_RUN

1.1 제작·검사 기준

- 단위: mm · 제3각법 · 축척: NTS(기입 치수 우선)
- 재료/구성: mica/NiCr heater and MI thermocouple
- 핵심 치수/공차: probe insertion ≥ 12 mm; heater-to-polymer path metal-only; shield clearance ≥ 12 mm
- 일반공차: ISO 2768-m. 개별 부품은 exports/final/manufacturing 도면과 interface_catalog.csv를 함께 검사한다.
- Source: adc8bec89d946f696c179d79b1af2c401189ab35 · 물리 검증: NOT_RUN